CLASS 7 — MATHS · FORMULA SHEET

Chapter 2: **Arithmetic Expressions** | Quick revision, one page

Order of operations (BODMAS)

- Brackets first, then multiplication and division (left to right), then addition and subtraction (left to right).
- Example: $6 + 2 \times (9 - 4) = 6 + 2 \times 5 = 16$. $12 \div 3 \times 2 = 4 \times 2 = 8$.

Properties

- Commutative: $a + b = b + a$, $a \times b = b \times a$ (not true for subtraction and division).
- Associative: $(a + b) + c = a + (b + c)$; $(a \times b) \times c = a \times (b \times c)$.
- Distributive: $a \times (b + c) = a \times b + a \times c$ and $a \times (b - c) = a \times b - a \times c$.
- Subtraction: $a - (b - c) \neq (a - b) - c$ in general.

Quick calculations

- $a \times 99 = a \times 100 - a$, $a \times 101 = a \times 100 + a$, $a \times 9 = a \times 10 - a$.

Forming expressions

- 'Sum of a and b' $\rightarrow a + b$; 'product' $\rightarrow a \times b$; 'difference' $\rightarrow a - b$; 'increased by' $\rightarrow +$; 'decreased by' $\rightarrow -$.
- Two expressions are equal if they have the same value; check by evaluating both.

MASTERMIND